Benchmarking six deep learning reconstruction models for sparse view photoacoustic tomography

[1]Yash Thakre, [2]Mridul Verma, [2]Partha Pratim Pal, [2,*]Renu John

[1]Department of Biotechnology, National Institute of Technology Warangal, Telangana, India

[2]Department of Biomedical Engineering, Indian Institute of Technology Hyderabad, Telangana, India

*renujohn@bme.iith.ac.in

Photoacoustic Tomography (PAT) combines optical contrast with ultrasonic resolution for biomedical imaging. Sparse-view acquisition produces severe reconstruction artifacts when conventional methods are used. Limited-view acquisition introduces severe artifacts when conventional Filtered Back-Projection (FBP) is used, motivating deep-learning-based reconstruction methods capable of restoring fine structural details.

We compare six representative reconstruction models using three different strategies. U-Net and FD-UNet operate on reconstructed images, with FD-UNet employing dense skip connections for improved feature propagation. Y-Net, FD-YNet, and AS-Net jointly utilize reconstructed images and raw acoustic measurements, while AS-Net incorporates attention-based feature fusion. DeepMB adopts a model-based projection framework that reconstructs directly from measurement data while preserving the underlying physical information. All models were trained and evaluated on the OADAT dataset using the 128-element sensor configuration, ensuring a consistent experimental setup for fair performance comparison.

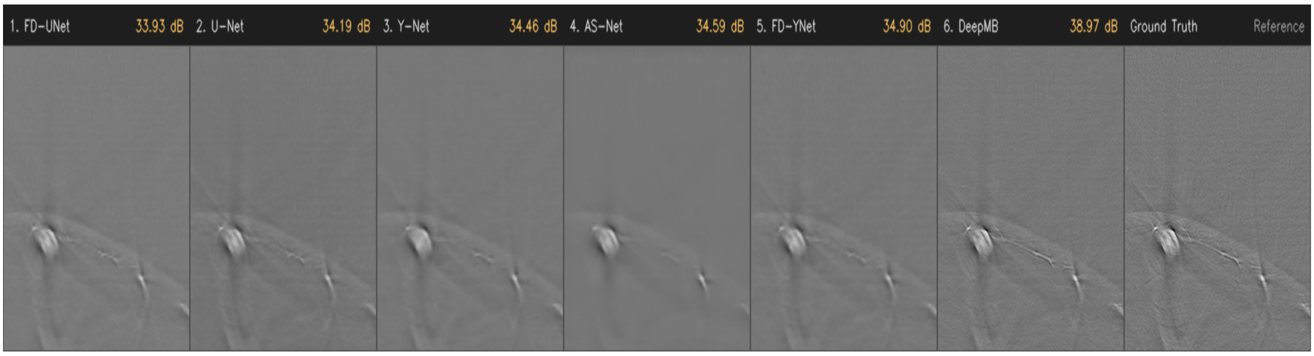

Fig. 1. *Visual comparison of Ground Truth and the six reconstruction methods.*

Model	PSNR (dB)	SSIM
DeepMB	38.97	0.9154
AS-Net	36.86	0.8901
FD-YNet	34.90	0.8291
Y-Net	34.46	0.8131
U-Net	34.19	0.8426
FD-UNet	33.93	0.8411

Table 1 summarizes the quantitative comparison. DeepMB achieves the highest PSNR by preserving measurement-domain information throughout the reconstruction pipeline. FD-YNet, AS-Net, and Y-Net achieve competitive intermediate performance by utilizing dual-input feature fusion and joint measurement-domain learning. Conversely, U-Net records a lower PSNR, experiencing smoothing that can affect fine structural details. The visual comparison in Fig. 1 supports these quantitative findings.

The results indicate that incorporating imaging physics into the reconstruction pipeline substantially improves sparse-view PAT performance. Physics-guided learning preserves structural information more effectively than purely image-domain enhancement, while dense feature reuse offers an efficient alternative with competitive reconstruction quality. Future work will focus on improving dual-input architectures through multi-scale feature learning and better measurement-domain fusion.

References

[1] Ko-Tsung Hsu, Steven Guan, and Parag V. Chitnis, “Comparing deep learning frameworks for photoacoustic tomography image reconstruction,” *Photoacoustics*, vol. 23, p. 100271, 2021.

[2] Mengjie Guo, Hengrong Lan, Changchun Yang, and Fei Gao, “AS-Net: Fast Photoacoustic Reconstruction with Multi-feature Fusion from Sparse Data,” *IEEE Transactions on Computational Imaging*, vol. 7, pp. 1180–1192, 2021.

[3] Christoph Dehner, Guillaume Zahnd, Vasilis Ntziachristos, and Dominik Jüstel, “DeepMB: Deep neural network for real-time optoacoustic image reconstruction with adjustable speed of sound,” *Medical Image Analysis*, vol. 84, p. 102715, 2023.

[4] F. Ozdemir, B. Lafci, X. L. Deán-Ben, D. Razansky, and F. Perez-Cruz, “OADAT: Experimental and Synthetic Clinical Optoacoustic Data for Standardized Image Processing,” *Transactions on Machine Learning Research (TMLR)*, 2023.